2. Factorization

Factorization of algebraic expression

2.1 Trinomials

Examples for trinomials

$x^2 - 5x - 6$, $2x^3 - 5x^2 - 3x$

$3x^2 - 5xy - 2y^2$

Example 1

Factorise : $x^2 - 5x - 6$

$= x^2 - 6x + x - 6$

$= x(x-6) + 1(x - 6)$

$= (x - 6)(x + 1)$

Example 2

Factorise : $2x^3 - 5x^2 - 3x$

$= x[2x^2 - 5x - 3]$

$= x[2x^2 - 6x + x - 3]$

$= x[2x(x - 3) + 1(x - 3)]$

$= x[(x - 3)(2x + 1)]$

$= x(x - 3)(2x + 1)$

Example 3

Factorise : $3x^2 - 4xy - 4y^2$

$= 3x^2 - 6xy + 2xy - 4y^2$

$= 3x(x - 2y) + 2y(x - 2y)$

$= (x - 2y)(3x + 2y)$

Example 4

Factorise : $2(x + 3)^2 - 7(x + 3) - 4$

Let $x + 3 = a$

$= 2a^2 - 7a - 4$

$= 2a^2 - 8a + a - 4$

$= 2a(a - 4) + 1(a - 4)$

$= (2a + 1)(a - 4)$

$= [2(x + 3) + 1][x + 3 - 4]$

$= (2x + 7)(x - 1)$

Example 5

Factorise: $2(2a+b)^2 - 5(2a+b)(a-2b) - 3(a-2b)^2$

Let $x = 2a+b$ and $y = a-2b$

$= 2x^2 - 5xy - 3y^2$

$= 2x^2 - 6xy + xy - 3y^2$

$= 2x(x-3y) + y(x-3y)$

$= (x-3y)(2x+y)$

$= [(2a+b) - 3(a-2b)]\ [2(2a+b) + (a-2b)]$

$= (7b-a)5a = 5a(7b-a)$

2.2 Difference of squares.

$a^2 - b^2$

$a^2 - b^2 = a^2 - ab + ab - b^2$

$= a(a-b) + b(a-b)$

$= (a-b)(a+b)$

$$\boxed{a^2 - b^2 = (a-b)(a+b)}$$

Example 1

Factorise: $a^3b - ab^3$

$= ab(a^2 - b^2)$

$= ab(a-b)(a+b)$

Example 2

Factorise: $x^4 - 1$

$= (x^2)^2 - 1^2$

$= (x^2 - 1)(x^2 + 1)$

$= (x-1)(x+1)(x^2+1)$

Example 3

Factorise: $a^4 + 4b^4$

$a^4 + 4b^4 = a^4 + 4a^2b^2 + 4b^4 - 4a^2b^2$

$= (a^2 + 2b^2)^2 - (2ab)^2$

$= (a^2 + 2b^2 - 2ab)(a^2 + 2ab + b^2)$

Example 4

Factorise: $1 - a^2 + 2ab - b^2$

$= 1 - (a^2 - 2aab + b^2)$

$= 1^2 - (a-b)^2$

$= [1 - (a-b)]\ [1 + (a-b)]$

$= (1 - a + b)(1 + a - b)$

3. Factorising $a^3 + b^3$ and $a^3 - b^3$

Consider the product $(a + b)(a^2 - ab + b^2)$

$(a + b)(a^2 - ab + b^2)$

$= a(a^2 - ab + b^2) + b(a^2 - ab + b^2)$

$= a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$

$= a^3 + b^3$

Therefore $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Consider the product $(a - b)(a^2 + ab + b^2)$

$(a - b)(a^2 + ab + b^2) = a(a^2 + ab + b^2) - b(a^2 + ab + b^2)$

$= a^3 + a^2b + ab^2 - a^2b - ab^2 - b^3$

$= a^3 - b^3$

Therefore $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

$$\boxed{\begin{aligned} a^3 + b^3 &= (a + b)(a^2 - ab + b^2) \\ a^3 - b^3 &= (a - b)(a^2 + ab + b^2) \end{aligned}}$$

Example 1

Factorise: $81x^3 - 3y^3$

$= 3[27x^3 - y^3)$

$= 3[(3x)^3 - y^3]$

$= 3(3x - y)[(3x)^2 + 3x \times y + y^2]$

$= 3(3x - y)(9x^2 + 3xy + y^2)$

Example 2

Factorise: $x^3 + \frac{1}{x^3}$

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)\left(x^2 - x \times \frac{1}{x} + \left(\frac{1}{x}\right)^2\right)$$

$$= \left(x + \frac{1}{x}\right)\left(x^2 - 1 + \frac{1}{x^2}\right)$$

Example 3

Factorise: $x^3 - \frac{1}{x^3}$

$$x^3 - \frac{1}{x^3} = \left(x - \frac{1}{x}\right)\left[x^2 + x \times \frac{1}{x} + \left(\frac{1}{x}\right)^2\right]$$

$$= \left(x - \frac{1}{x}\right)\left(x^2 + 1 + \frac{1}{x^2}\right)$$

Example 4

Factorise: $8a^3 + (b + c)^3$

$$8a^3 + (b + c)^3 = (2a)^3 + (b + c)^3$$
$$= [\,2a + (b + c)]\,[\,(2a)^2 - 2a\,(b + c) + (b + c)^2\,]$$
$$= (\,2a + b + c)\,(4a^2 - 2ab - 2ac + b^2 + 2bc + c^2\,)$$
$$= (\,2a + b + c)\,(4a^2 + b^2 + c^2 - 2ab - 2ac + 2bc)$$

Example 5

Factorise: $a^3 - 27(b - c)^3$

$$a^3 - 27(b - c)^3 = a^3 - \{3(b - c)\}^3$$
$$= [\,a - 3(b - c)]\,[\,a^2 + 3a(b - c) + 9(b - c)^2\,]$$
$$= (\,a - 3b + 3c)\,(\,a^2 + 9b^2 + 9c^2 + 3ab - 3ac - 18bc)$$

Example 6

(i) **Factorise:** $(a + b)^3 + c^3$

(ii) Write the expansion of $(a + b)^3$ and show that $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$

(iii) **Using the above results factorise** $a^3 + b^3 + c^3 - 3abc$

(i) $(a + b)^3 + c^3 = [(a + b) + c\,][(a + b)^2 - c(a + b) + c^2\,]$
$$= (a + b + c)\,(a^2 + 2ab + b^2 - ac - bc + c^2)$$
$$= (\,a + b + c)\,(a^2 + b^2 + c^2 + 2ab - ac - bc)$$

(ii) $(a + b)^3 = a^3 + 3a^2 b + 3ab^2 + b^3$

$(a + b)^3 - 3a^2 b - 3ab^2 = a^3 + b^3$

$(a + b)^3 - 3ab(a + b) = a^3 + b^3$

$a^3 + b^3 = (a + b)^3 - 3ab\,(\,a + b)$

(iii) $a^3 + b^3 + c^3 - 3abc$

$=(a+b)^3 - 3ab(a+b) + c^3 - 3abc$; **[from (ii)]**

$=(a+b)^3 + c^3 - 3ab(a+b) - 3bc$

$=[(a+b)+c][(a+b)^2 - c(a+b) + c^2\,] - 3ab[a+b+c]$; **[from (i)]**

$=(a + b + c)(a^2 + b^2 + c^2 + 2ab - ac - bc) - 3ab(a + b + c)$

$=(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Exercise 2.1

Factorise:

1	$x^2 - x - 6$	2	$x^2 + 4x - 96$
3	$x^2 + 5x - 6$	4	$x^2 - 4x - 12$
5	$x^2 + x - 42$	6	$x^2 - 9x + 18$
7	$2x^2 + 5x + 3$	8	$2x^2 - 5x + 3$
9	$2x^2 + 5x - 3$	10	$2x^2 - 5x - 3$
11	$10 - 7x - 12x^2$	12.	$15 + x - 2x^2$
13.	$18x^2 - 33x - 216$	14	$6x^2 - 55x + 126$
15	$2x^2 - 5xy + 3y^2$	16.	$6x^2 - 5xy - 6y^2$
17.	$4x^2 + 8xy + 3y^2$	18.	$2a^2 - 27ab + 13b^2$
19.	$40x^2y^2 + 49xy - 24$	20	$32x^2 - 36xy - 35y^2$
21.	$24a^3 - 17a^2b - 20\,ab^2$	22.	$18a^3 - 3a^2b - 10ab^2$
23.	$(a^2 - 3a)^2 - 38\,(a^2 - 3a) - 80$	24	$(a + b + c)^2 - 3(a + b + c) - 28$
25	$2(x+y)^2 - 3(x+y) - 27$	26	$2(2x+y) - 5(2x+y)(x-2y) + 3(x-2y)^2$
27.	$x^2 + x - (a-1)(a-2)$	28	$x^2 - x - (a-1)(a-2)$
29	$x^2 - \left(a + \frac{1}{a}\right)x + 1$	30	$x^2 + 2ax + (a+b)(a-b)$
31	$ax^2 + (ab-1)x - b$	32	$x^2 + ax - (6a^2 - 5ab + b^2)$

33. $4(a^2 - b^2)^2 - 8ab(a^2 - b^2) - 5a^2b^2$

34. $10(a+2b)^2 + 21(a+2b)(2a-b) - 10(2a-b)^2$

35. $6(x + y)^2 - 5(x^2 - y^2) - 6(x-y)^2$

Exercise 2.2

Factorise:

1	$x^2 - 4y^2$	2.	$x^3 - x$	3	$x^2 - \frac{1}{x^2}$
4.	$x^5 - x$	5	$4 - 9a^2$	6.	$(a-4b)^2 - 9b^2$
7.	$16 - (a+b)^2$	8.	$9 - (a-b)^2$	9.	$12a^3 - 3ab^2$
10.	$1 - (a-b)^2$	11.	$1 - (a+b)^2$	12.	$x^2 - y^2 - x - y$
13	$x^2 - y^2 - x + y$	14	$x^2 - y^2 + x + y$	15	$x^2 - y^2 + x - y$
16	$a^2 - b^2 - 4a + 4b$	17.	$a^2 - b^2 - 4a + 4$	18	$ab + ac - (b+c)^2$
19.	$a(a+1) - b(b+1)$	20	$x^4 - 3x^2y^2 + y^4$	21.	$x^4 + x^2y^2 + y^4$
22.	$a^4 + 5a^2b^2 + 9b^4$	23.	$x^2 - 4xy + 4y^2 - z^2$	24.	$4a^2 + b^2 - x^2 + 4ab$
25	$x^4 + x^2 + 1$	26.	$4a^4 + 11a^2b^2 + 9b^4$		

Evaluate:

1	$100^2 - 99^2$	2.	$94^2 - 36$	3.	$12.38^2 - 7.62^2$
4.	$6.2^2 - 3.8^2$	5.	$100 \times 99 + 1$	6.	$11.7 \times 9.3 + 8.3 \times 9.3$
7.	$\sqrt{148 \times 140 + 16}$	8.	$319^2 - 318 \times 320$	9	$12.5^2 - 13 \times 12$
10	103×97				

Exercise 2.3

Factorise:

1	$a^3 + 8b^3$	2.	$27a^3 - b^3$	3.	$125a^3 - 64b^3$
4.	$8a^3b^3 - c^3$	5.	$x^3 + \frac{1}{x^3}$	6.	$x^3 - \frac{1}{x^3}$
7.	$\frac{1}{a^3} + \frac{1}{b^3}$	8.	$\frac{1}{a^3} - \frac{1}{b^3}$	9.	$a^3 + (b + c)^3$
10.	$a^3 + (b - c)^3$	11.	$a^3 - (b - c)^3$	12.	$8x^3 + (2y - x)^3$
13.	$(a + b)^3 + (a - b)^3$	14.	$(a + b)^3 - (a - b)^3$	15.	$8(a + b)^4 + (a + b)$
16	$x^6 - y^6$	17.	$x^6 + y^6$	18.	$x^6 - 27$

19. (a) Factorise $(a+b)^3 + c^3$

(b) Show that $a^3 + b^3 = (a + b)^3 - 3ab(a+b)$

(c) Using the results in (a) and (b), factorise $a^3 + b^3 + c^3 - 3abc$

Hence, factorise the following polynomials.

(i) $x^3 + y^3 - z^3 + 3xyz$

(ii) $8x^3 + y^3 + z^3 - 6xyz$

(iii) $x^3 + 8y^3 - 27z^3 + 18xyz$

(iv) $a^3 - 28b^3 - 9ab^2$

(v) $8a^3 + b^3 - 1 + 6ab$

20 **Show that**

(i) **if** $a = b + c$, **then** $a^3 - b^3 - c^3 = 3abc$

(ii) **if** $a + b + c = 0$, **then** $a^3 + b^3 + c^3 = 3abc$

(iii) **if** $z = 2x - 3y$, **then** $8x^3 - 27y^3 - x^3 = 18xyz$

21. **Given that** $x + y + z = 0$, **show that** $x^3 + y^3 + z^3 = 3xyz$

Hence, factorise the following algebraic polynomials.

(i) $(a-b)^3 + (b-c)^3 + (c-a)^3$

(ii) $(2x - 3y)^3 + (3y - 4z)^3 + 8(2z-x)^3$

(iii) $a^3(b-c)^3 + b^3(c-a)^3 + c^3(a-b)^3$

(iv) $(x - 3y)^3 + (3y - 4z)^3 + (4z - x)^3$